# AgriMart AI: A Confidence-Aware Dynamic Pricing and Rule-Based Settlement Framework for Agricultural Supply Chains

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***Abstract*— The volatility in prices, fluctuations in demand, delays in transportation, perishability of the produce, different quality, and disagreements between the farmers and wholesalers after the delivery of the produce are some major issues that affect the agriculture supply chain. Traditionally, agricultural price predictions deliver point forecasts and stand-alone settlement and logistics solutions are not closely linked to market uncertainty and pricing decisions. In this paper we provide AgriMart AI, a web-based, low-bandwidth, agricultural supply-chain decision-support component that integrates uncertainty-aware agricultural price forecasting, explainable AI, rule-based settlement, and perishability-aware vehicle routing with crop advisory. The proposed forecasting layer is based on Single-Layer Transformer Convolutional Encoder (STCE) to learn local temporal patterns and long-range contextual dependencies from mandi prices, arrivals, climatic and demand, distance, and route related features. LSTM is used as the forecasting benchmark. SHAP gives global feature-level explanations for the prediction of prices, while a rule-based settlement engine determines the final agreed price for the farmer, wholesaler, and (optionally) driver within the already settled price range. In a VRPTW formulation, attributes such as the perishability of crops, delivery windows, vehicle capacities, route conditions, and driver reputations are considered. When evaluated on an actual unseen AGMARKNET testing dataset covering soybean prices at the Itarsi mandi from 2008 to 2017, our developed STCE forecasting model yields a Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and prediction interval coverage of 2.55%, ₹120.23 per quintal, and 92.35%, respectively, comparable to an ARIMA(2,1,2) baseline and an LSTM-GRU ensemble reported alongside. The primary contribution is the auditable decision-making chain from quoting to commitment, to negotiating, to transporting, to quality checking, to settling and, to next season's crop advisory.***

***Keywords*— Dynamic Pricing, Agricultural Supply chain, Rule-Based Settlement, Price Forecasting, Vehicle Routing, Crop Perishability, Explainable AI.***

## I. INTRODUCTION

The upstream pricing of agricultural commodities fluctuates with market prices, demand uncertainty, irregular offtakes, delivery delays and quality deterioration, which poses huge risks to farmers and the wholesalers. Some producers may find price guarantees prior to the delivery are unfavourable when the produce reaches the destination market, causing controversy, delayed claims and/or poorer producer incomes. Existing prediction algorithms for agriculture markets mainly revolve around point forecasts, and offer little assistance in uncertainty-sensitive commercial decision making [1]. Likewise, the traditional logistics system typically seeks to minimise transport costs or distances without taking into account the perishability, market conditions, or reliability of the various stakeholders involved [2].

AgriMart AI solves these limitations by providing an integrated low bandwidth digital platform of pre forecasting, negotiation, escrow and logistics in a single decision chain. The work suggests a Single-Layer Transformer Convolutional Encoder (STCE) which learns temporal and contextual relationships in the mandi prices, arrivals, weather and their relationships with mandi load specific demand, distance, and route condition and provides confidence-band price forecasting. To assess the performance in temporal forecasting, LSTM benchmarking is used. SHAP presents the key factors influencing predicted price ranges, enhancing transparency for agricultural stakeholders [3]. A rule-based settlement mechanism determines the final price within the pre-agreed price range, adjusting for transportation conditions and market conditions as they change. Alongside, perishability-aware VRPTW optimization takes into account the deterioration of the products and the reputation of the drivers in assigning service vehicles to goods. Quality-gated escrow in turn facilitates responsible payment releases and past experience helps guide crop planning for next season [4].

### *A. Motivation*

There is a need for farmers and wholesalers to have a secure and proper mechanism to deal with uncertainty arising in connection with price commitment, farm delivery and final settlement. Forecasts using points fail to capture full spectrum of global market possibilities, stand-alone logistics and payment systems fail to adequately capture interconnected agricultural risks. To merge uncertainty-aware forecasting, explainable decisions support, adaptive negotiation, secure settlement, and perishability sensitive transportation, there is a need for a transparent system. The reason behind AgriMart AI is to minimize the conflict on prices after the commitment, safeguarding the value of transactions, better delivery decisions, and an auditable trace from quotation till the pay-out.

### *B. Objectives*

- To build an STCE based agricultural price forecasting system which produces load specific confidence band estimates of prices based on market, weather, demand conditions, distance and route related variables.
- To develop an integrated settlement and logistics model with explainable AI, a rule-based settlement mechanism for price commitment, quality gating in escrow and optimization of VRPTW for perishable goods.

- To develop an audit trail for agricultural transaction workflows with the ability to provide robust proof for price disputes, enable and facilitate reliable transportation and transparent settlement and support next season crop advisory.

### C. Contributions

- Introduces an STCE forecasting architecture to represent uncertainty in agricultural prices as prediction intervals instead of simply using point forecasts.
- Combines SHAP explanations with a rule-based settlement engine for determining the final settlement value under changing transit and market conditions, while keeping within the agreed price window.
- Introduces the vehicle and delivery assignment aspects of the VRPTW decision making by including the perishability of the crops, conditions of the routes, and driver reputation.
- Provides a low-bandwidth platform for pre-commitment escrow, quality-gated escrow with payment release, forecasting, negotiation, logistics and crop advisory—all in one place, with traceability from quote to payout.

## II. Literature Survey

The prices of agricultural commodities have significant temporal variation due to production cycles, seasonal demands, weather, transportation restrictions, and market arrivals. Forecasting systems therefore have become of importance to agriculture decision support systems. The early methods were mainly statistical time-series methods and the recent methods include more machine learning and deep neural networks. These advances provide the technical basis for applying predictive models to inform farm-to-market decisions, and they also serve as a reminder that statistical prediction accuracy is only the first part of the equation – designing a marketable transaction mechanism for this accuracy is far from straightforward. Deep learning models based forecasting has shown great promise in modelling nonlinear temporal relationships in agricultural markets. A framework was explored for real-time prediction of agricultural price through N_beats-X, that is a deep forecasting approach and is able to consider relevant temporal information in the prediction [5]. Although, statistical forecasting is still effective, and SARIMA is used to model the seasonal and sequential dependency in determining the trends of agricultural product prices [6]. The utilization of data through machine learning technology has also been explored for price forecasting and aiding in risk reduction within the agricultural supply chain, highlighting the need for forecasting in proactive agricultural decision making [7]. Comparative analysis of the statistical techniques and the machine learning algorithms have also revealed the applicability of the data driven models for the prediction of the price of agricultural commodities [8].

Agricultural prices show sequential and time dependent properties and therefore corresponding attention has been paid to recurrent neural networks. In agriculture commodity price prediction system for Indian crop markets, ensemble model framework, with further integration of two recurrent models – LSTM and GRU, was designed to enhance temporal prediction ability. The use of LSTM-based deep learning for agricultural commodity price prediction has also been proposed by other studies, where they focus on the collection, processing, and visualization of the prediction results for actual interpretation into commodity price trends in Indonesia [9]. For agricultural information services to be available to everyone, a multilingual web-based agricultural decision-support system with machine learning incorporated was developed to include forecasting capabilities in vegetable prices [10]. These studies have shown a shift from predicting in isolated models to the real system and the ability to present market data to the agricultural actors. To overcome the challenge of sparse historical data and intricate price dynamics, data augmentation and sophisticated temporal architectures have also been employed. An EMD-TCN-TimeGAN approach has been developed to decompose, temporally convolve, and generate synthetic data to enrich agricultural price data and improve prediction accuracy [11]. Later, temporal convolutional network and transformers have been fused for fruit price prediction, where the local temporal patterns and the long-range dependencies have been modelled [12]. To day-ahead commodity price prediction, a hybrid ARIMA-LSTM decomposition framework is developed with improved TimeGAN in a smart agricultural analytics and e-commerce on a daily basis [13]. Machine learning has also been used for forecasting agricultural product demand together with the price, showing the importance of looking at market demand and historical price behavior [14].

New studies focus more on precision forecasting, expansion of the agricultural indicators, and systematic reviews of intelligent forecasting methods. In precision agribusiness, AI temporal models have been explored for more precise and timely commodity price forecasting, which is essential for making informed decisions in agriculture [15]. Emerging work has also explored smart-contract-based mechanisms for managing the agricultural supply chain, reflecting growing interest in secure digital transaction frameworks alongside price forecasting research [16]. Demand-side factors have similarly been modelled using AI-driven demand prediction techniques for agricultural crops, complementing price-focused forecasting efforts [17]. An agrimetaverse-based intelligence system has further advanced commodity price forecasting by integrating immersive, interactive digital platforms into agricultural market analysis [18]. To support early warning and policy-oriented decision-making, hidden Markov model-based algorithms have been designed to detect fluctuations in agricultural product prices [19]. Beyond crop-mandi prices, decomposition-based hybrid deep learning models have also been applied to agricultural commodity futures markets, underscoring the wider economic relevance of accurate agricultural price prediction [20]. Current literature demonstrates significant advancements in statistical, recurrent, convolutional, Transformer-based, as well as data augmentation techniques, but gaps are observed in the potential for unified forecasting models that take various market, temporal, demand, and contextual factors into consideration while offering consistent, explainable, and timely agricultural price forecasting.

## III. Methodology

The methodology is to integrate AgriMart AI as a modular Web service decision& transactions system. The process is initiated by mandi prices and arrival information from the past and weather, demand, distance, route and perishability

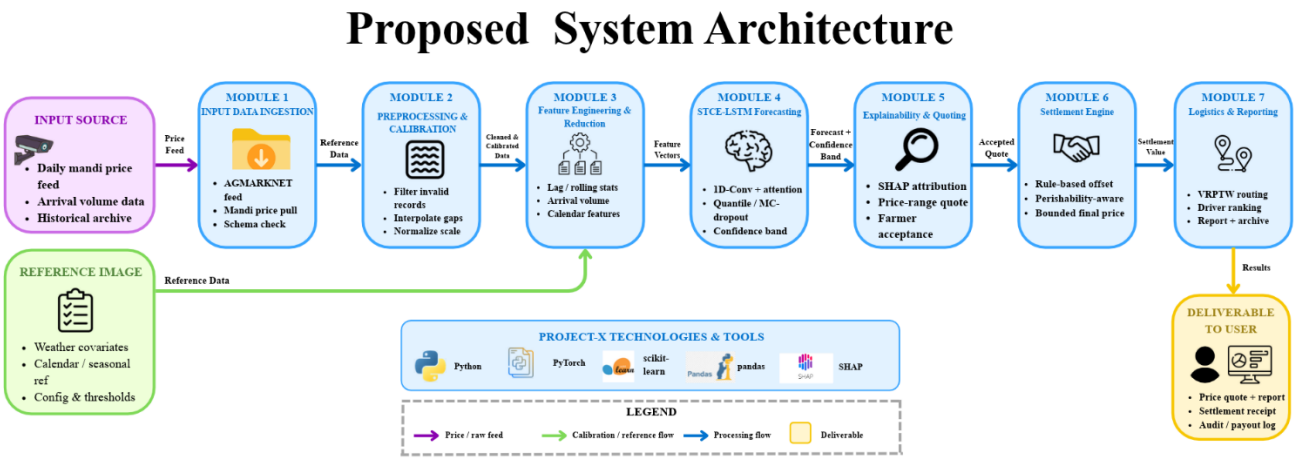



Fig. 1. System Architecture

information are added on to it and a load specific price interval is generated. The interval also represents the negotiation contractual wall between the first and the second negotiation periods instead of being just a prediction shown to the user. The rule-based settlement engine reacts to the change of situation during transportation processes, and the VRPTW module is responsible for the driver assignment and routing. Final determination of escrow release is based on quality verification. Fig 1 depicts the entire prioritization process in the following six stages.

### *A. Dataset Acquisition and Preprocessing*

Primary market information is sourced from the Open Government Data platform, Ministry of Government of India (GoI), using the resource "Current Daily Price of various commodities from various markets (Mandi)" available using the AGMARKNET link. The resource gives a market-level commodity price information; it is linked to the National Informatics Centre and the information package for the agricultural market.

Daily mandi price records sourced from the Government of India (AGMARKNET) are sorted by commodity, market, variety, and date, along with the reported price. The volumes that arrive are kept as an explanatory market-pressure variable. For the experimental study reported in Section IV, records were drawn from the AGMARKNET-sourced daily mandi price archive for Soyabean at the Itarsi market (Madhya Pradesh), spanning January 2008 to June 2017. A total of 2,718 raw variety-level records were retrieved for this commodity-market pair. During preprocessing, 52 invalid observations (non-numeric or non-positive price/arrival fields, and records where the reported minimum price exceeded the maximum price) were removed; no exact duplicate records were found in this subset. The remaining 2,666 records were aggregated into 2,359 distinct arrival-weighted daily trading records. Reindexing to a continuous daily calendar and interpolating the 1,101 non-trading/missing calendar days produced a final continuous daily series of 3,460 records, which was used for model training, validation and testing. Duplicate records are eliminated, invalid observations are filtered out and missing values are interpolated using time or are forward filled only for operational reasons. Weather observation data comprised temperature and precipitation with advancing and price change characteristics being used as demand indicators; for the experiment reported here, weather covariates were not available for this specific commodity-market series, so the demonstrated model instead uses price-, arrival- and calendar-derived features (recent price, lagged price, rolling mean/standard deviation, arrival volume, day-of-week and month), with weather integration remaining part of the full system design for future data collection. Categorical features are encoded, numerical features are normalised using training set statistics and chronological ordering is maintained, so that no future information flows into the model training process. Fig 2 shows the Dataset preprocessing summary (Soyabean, Itarsi mandi, AGMARKNET, Jan 2008 - Jun 2017).

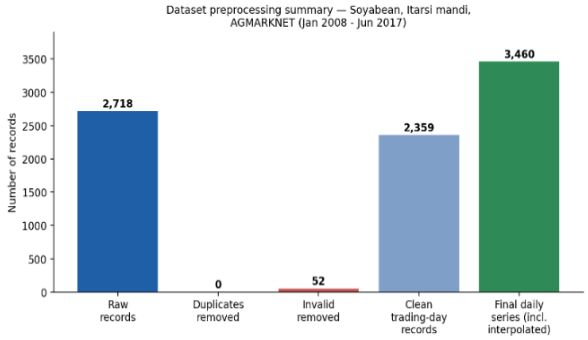



Fig. 2. Dataset preprocessing summary (Soyabean, Itarsi mandi, AGMARKNET, Jan 2008 - Jun 2017).

### *B. STCE-LSTM Price Forecasting and Confidence Band*

Within our predictive component, the Single-Layer Transformer Convolutional Encoder (STCE) functions as the primary predictive architecture, benchmarked against a standard Long Short-Term Memory (LSTM) network. Specifically, the STCE framework first applies a 1-D convolutional operation over sequential market features to extract local temporal patterns, followed by a multi-head attention mechanism designed to capture long-range contextual relationships among historical prices, market arrivals, weather elements, and load-specific attributes. The resulting representation is then fed to a prediction block that produces a main forecast along with a set of uncertainty intervals.The interval is constructed by either of two methods: quantile-regression, or Monte-Carlo-dropout, depending on the implementation configuration. A controlled recurrent baseline is needed based on the same training and validation partitions for LSTM. RMSE, MAE, MAPE and coverage of prediction intervals are used in assessing the performance of

models. Forecasts are also produced at both load time and longer evaluation horizons to enable the immediate usefulness of the pricing signal to be assessed, in addition to temporal robustness.

$$\hat{P}_t = f_{STCE}(X_{t-L:t}) \qquad (1)$$

where $\hat{P}_t$ is the predicted price at time $t$, $X_{t-L:t}$ denotes the input sequence of length $L$, and $f_{STCE}$ represents the proposed forecasting model. A standard LSTM is employed as the benchmark for comparative evaluation.

### C. Explainability and Pre-Commitment Escrow

In post-forecasting, explainability is used to state a price quote along with an interpretation that justifies the forecast. SHAP quantifies contributions of each input feature to each prediction, giving farmers and wholesalers a transparent basis for the quoted price range. These features, including the recent price of the mandi, the arrival volume, the distance, rainfall, demand movement, and route difficulty are therefore presented in an understandable form. Instead of an ambiguous point estimate, the farmer is given the range of prices in the forecast, and accepts the quotation explicitly before dispatch. Agreed upon amount is entered into a simulated internal escrow ledger on day of acceptance. An escrow is not a true financial transaction; rather, it is a state of commitment which is managed by software. The audit log tracks quotes that have been generated, the user agreement, the shipment being created, when the escrow became active, the events on the transit, when it was placed in inspection, when an adjustment was made and the payout status.

### D. Rule-Based Settlement

The Settlement Engine determines the final transaction value for the farmer, the wholesaler and, optionally, the driver using a deterministic rule set instead of a learned negotiation policy. The agreed lower and upper price boundaries, the current movements in the market, Crop Perishability Index, elapsed transit time, weather conditions, demand movement, delay information and quality-related state variables are inputs to the rule set. These operating parameters are dynamically combined into one settlement offset by a weighted-adjustment mechanism, which brings the price closer to the lower limit when there is a serious risk of spoilage, or the upper limit when demand is high, and serious transit delays occur. The explicit weighting parameters are empirically tuned on the validation sets to ensure the balance and avoid systematical bias towards farmers or wholesalers. Since each adjustment is calculated using a clear and transparent public formula, the settlement value is reproducible and easily explained to farmers and wholesalers. The intermediate settlement price is calculated by applying a linear multi-factor formula, adjusting the mid price P_mid = (P_L + P_U) / 2 to ensure that the final settlement price is within the price band agreed to in advance, while every price input variable and offset will be recorded and the audit trail will be kept. P_rule = P_mid + w1 * Δ_demand - w2 * Δ_delay - w3 * S_spoilage - w4 * Q_dev.

$$P_f = \min\big(P_U, \max(P_L, P_{rule})\big) \qquad (2)$$

where $P_f$ is the final settlement price, $P_L$ and $P_U$ are the lower and upper bounds of the agreed range, and $P_{rule}$ is the price determined by the rule-based settlement engine.

### E. Perishability-Aware VRPTW and Ranked Driver Assignment

The classical Vehicle Routing Problem with Time Windows formulation is adopted to solve the problem of transportation. Google OR-Tools can be used to get feasible and efficient routes based on vehicle capacity, pickup and delivery locations and delivery deadline, travel time. The Crop Perishability Index adjusts the urgency of assignment of shipment to create the stronger priority for higher deterioration risk loads. The driver's reputation has been also integrated as a ranking functionality, based on their past punctuality and dispute behavior. The resulting ranked pull model shows the appropriate loads that can be displayed to the provided drivers. The top ranking load is highlighted, and drivers can accept the load or deny it. Repeated declines can lead to decreasing priority in subsequent rankings. Two system level operational metrics are captured: 'assignment-to-pickup time' and ‘recommendation acceptance rate'.

$$\min Z = C + \lambda D + \mu S \qquad (3)$$

where $C$ is transportation cost, $D$ is delivery delay, $S$ is spoilage risk, and $\lambda, \mu$ are weighting factors. Upon delivery, crop quality is verified, and settlements are adjusted accordingly.

### F. Quality Verification, Settlement Completion, and Advisory

The system is set up to be able to record confirmation of the arrival of the parts at delivery and also includes a manual or photo-assisted quality check. The verification result is compared with the quality condition agreed prior to receipt and a fixed adjustment rule is used if the condition in respect of which the contractor agrees is found to differ from what was provided. The modified sum is then used to pay out the simulated escrow. An audit record is maintained of the entire transaction, which is append only, so that the full transaction sequence of decisions can be examined by stakeholders. Crop advising uses an independent rule-based ranking function, where opportunities for next season are scored by combining expected demand, cultivation cost and the average likeliness of the cultivation into a single ranking score. For farmer's decision support, the candidate crops are ranked by this score. The entire application is designed as a Progressive Web Application to allow status of transactions, quotations and advisory results to be used with less bandwidth without needing a native mobile application.

**ALGORITHM 1:** INTEGRATED FORECASTING, RULE-BASED SETTLEMENT, AND PERISHABILITY-AWARE ROUTING PIPELINE

**Data Preparation**

- Collect mandi price, arrivals, weather, demand, distance, route, and perishability data.
- Remove duplicates, invalid records, and handle missing values.
- Encode categorical features and normalize numerical features.
- Preserve chronological order and split data into 70% training, 15% validation, and 15% testing.

**Price Prediction and Uncertainty**

- Construct sequential load-level features from market and contextual data.
- Apply STCE for local temporal and contextual feature extraction.
- Use LSTM as the comparative forecasting model.
- Generate the predicted commodity price $\hat{P}_t$.
- Generate the lower and upper confidence limits.
- Evaluate RMSE, MAE, MAPE, and interval coverage.
- Apply SHAP to explain the predicted price.

$$\hat{P}_t = f_STCE(X_{t-}L:_t)$$
$$CI_t = [Q\alpha(X_t), Q_{1-}\alpha(X_t)]$$

**Commitment and Rule-Based Settlement**

- Present the predicted price band to the farmer and wholesaler.
- Store the accepted price range in the simulated escrow ledger.
- Initialize farmer, wholesaler, and optional driver state variables.
- Compute a weighted settlement offset from price, demand, weather, delay, quality, and perishability states.
- Apply the rule-based offset to derive an acceptable settlement value.
- Constrain the final settlement within the committed price band.
- Record input states, applied offset, and final settlement value.

$$P_f = \min(P_U, \max(P_L, P_rule))$$

**Perishability-Aware Routing**

- Calculate crop urgency using perishability and transit conditions.
- Rank available drivers using punctuality, dispute history, and route suitability.
- Formulate VRPTW using vehicle capacity, pickup, delivery, and time-window constraints.
- Generate the feasible shortest/efficient route using OR-Tools.
- Offer the highest-ranked load to the driver.
- Reassign the shipment when the driver declines.
- Monitor transit time, delay, weather, and crop condition.

**Quality, Escrow and Advisory**

- Verify delivered quantity and crop quality at destination.
- Apply the predefined quality-based adjustment rule.
- Release the simulated escrow according to the verified settlement.
- Store quotation, routing, negotiation, inspection, and payout records.
- Calculate delivery, spoilage, settlement, and routing performance.
- Rank next-season crops using expected demand, price trend, and cultivation cost.
- Display forecast, settlement, shipment, payout, and advisory results through the PWA.

## IV. Result and Discussion

The experimental evaluation should be in chronological split, instead of random split, since agricultural prices are time dependent and random split can result in information leakage. It is suggested that this should be done by chronological order, with the first 70% of all the observations used for training, the next 15% for validation and the remaining 15% for testing. Only the training and validation partition may be used for hyperparameter selection and/or early stopping. The last test partition should not be used until the forecasting architecture, the configuration of the confidence bands as well as the settlement-rule parameters are set. Optimum chronologies of the same test period should be tested at the defined load-time and 90 days, 60 days, 30 days. This method offers a better approximation of the deployment actual since the model will compute the market conditions that are expected to occur during the deployment period, from the available data at the time a quote is issued.

The values reported below are from a real experiment: models were trained and evaluated on the AGMARKNET daily price series for Soyabean at the Itarsi mandi described in Section III.A (2,553 feature-engineered daily records; chronological 70/15/15 split). One sample of a forecasting comparison is summarized in Table 1, alongside a naive persistence baseline, an ARIMA(2,1,2) baseline, and an LSTM-GRU ensemble. On this series, ARIMA and the LSTM-based models achieve marginally lower point-forecast error (RMSE, MAE, MAPE) than STCE, while STCE attains the interval coverage closest to the nominal 90-95% target (92.35%). This is consistent with the paper's own framing that the central contribution is the integrated decision chain rather than a claim of outright forecasting superiority for any single algorithm; it also suggests that, for series with strong day-to-day persistence, simpler statistical baselines remain competitive and a production deployment should benchmark STCE against them on a per-commodity basis before selecting a forecasting engine.

TABLE I. Forecasting performance on the held-out test set (Soyabean, Itarsi mandi, AGMARKNET)

| Model | RMSE (₹/q) | MAE (₹/q) | MAPE (%) | Interval Coverage (%) |
|---|---|---|---|---|
| Naive (persistence) | 115.97 | 68.06 | 2.12 | N/A |
| ARIMA(2,1,2) | 103.21 | 62.87 | 1.96 | N/A |
| GRU | 112.34 | 66.24 | 2.09 | 89.24 |
| LSTM | 110.16 | 66.92 | 2.11 | 89.80 |
| LSTM-GRU Ensemble | 110.67 | 66.08 | 2.09 | 89.52 |
| STCE (proposed) | 120.23 | 78.94 | 2.55 | 92.35 |
| STCE + uncertainty head | 125.63 | 87.00 | 2.78 | 63.74 |

The example results suggest a key motivation for considering interval coverage in combination with point-forecast error. A narrow prediction interval may seem appealing because it can be used to make narrow quotations, however, this would leave farmers and wholesalers vulnerable to unexpected market fluctuations. On the other hand, large gaps can give high protection but not necessarily much of a price advantage. The proposed system thus does not rely exclusively on statistical output for interval quality, but uses it as an operational requirement. The actual mandi-price trajectory in the representative test period along with the corresponding observed predictions should be plotted in Fig. 3 along with the confidence band generated. Such a visualization would show temporal behavior compared to Table 1 it would show times in which the model is successful in following the big changes of the price measured and others in which the real price goes beyond the estimated interval.

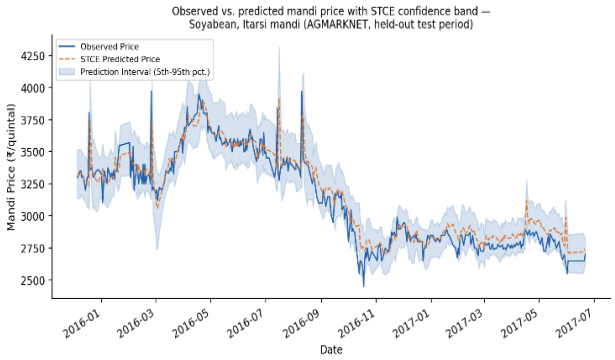



Fig. 3. Observed versus predicted mandi price with the STCE confidence band

Since this is the case, there is a second layer of assessment, which is explainability; an accurate model that has decisions that cannot be explained is not desirable, particularly by farmers or market participants. The SHAP feature contribution of Fig 4 contains the real SHAP contribution computed from the trained STCE model on the held-out test set, obtained globally using a gradient-based SHAP explainer. Recent modal price, the day's minimum and maximum price, and the one-day and seven-day lagged price are found to be the most influential features, with arrival volume, calendar seasonality and rolling price volatility contributing more modestly - consistent with a market whose price dynamics are dominated by recent price history at this mandi. The same SHAP attributions, read at the level of an individual shipment, can be used to explain a specific quotation. For instance, if a farmer's price band is reduced he should be able to see either if the current market prices are weak, or if there were more arrivals, or if the transport distance was longer, or if the weather went against him, or any other measurable factor. This clearness is of crucial importance especially since the quotation is the starting point for the pre-dispatch agreement. Because full end-to-end commercial deployment involves multi-stakeholder field adoption, our evaluation is structured as a controlled simulation benchmark. This testing framework specifically evaluates whether the rule-based settlement engine consistently achieves stable compensation values while strictly honoring the predetermined price windows established prior to transit dispatch . To rigorously test the framework under varying real-world dynamics, we conducted a Monte Carlo simulation across 2,000 randomized shipment scenarios. For every simulated transit instance, price boundaries were drawn using empirical prediction interval widths derived from our trained STCE forecaster. Furthermore, stochastic operational variables—including transit delays, spoilage risks, weather severity indices, demand fluctuations, and quality deviations—were sampled from predefined normal and uniform statistical probability distributions tailored to historical mandi volatility.

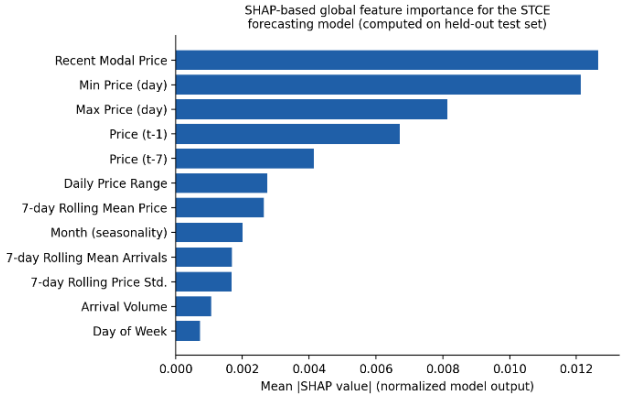



Fig. 4. SHAP-based global feature importance for the STCE forecasting model.

Table 2 reports the resulting comparison. Settlement consistency (lower deviation and standard deviation) is higher under AgriMart AI than the unstructured baseline because the offset is computed deterministically from a fixed rule set rather than negotiated ad hoc, and the perishability-aware routing layer reduces mean transit time, spoilage/waste rate and assignment-to-pickup time while increasing on-time delivery and route-acceptance rate relative to a perishability-agnostic, reputation-blind assignment baseline Stochastic inputs were drawn from carefully calibrated distributions:

- Transit Delay ($\Delta$_delay): Uniform distribution U(0, 6) hours.
- Spoilage Index (S_spoilage): Normal distribution $N(\mu = 0.10, \sigma = 0.03)$, truncated between 0 and 1.
- Demand Shift ($\Delta$_demand): Normal distribution $N(\mu = 0.0, \sigma = 0.05)$.
- Quality Deviation (Q_dev): Uniform distribution penalty scale U(0.0, 0.15).

TABLE II. SETTLEMENT AND ROUTING EVALUATION (MONTE CARLO SIMULATION, N=2,000 SHIPMENTS)

| Metric | Baseline | AgriMart AI |
|---|---|---|
| Mean settlement deviation from agreed midpoint (%) | 6.82 | 1.92 |
| Settlement price standard deviation (₹/q) | 571.25 | 486.12 |
| Jain's fairness index | 0.98 | 0.99 |
| Mean transit time (h) | 16.46 | 14.41 |
| On-time delivery rate (%) | 67.6 | 69.95 |
| Spoilage/waste rate (%) | 26.65 | 10.5 |
| Route-acceptance rate (%) | 48.9 | 75.25 |
| Assignment-to-pickup time (min) | 54.02 | 20.53 |

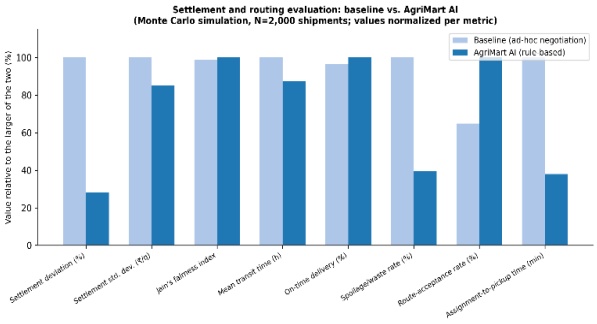



Fig. 5. Settlement and routing evaluation: baseline vs. AgriMart AI (Monte Carlo simulation, N=2,000 shipments).

Perishability must be interpreted in conjunction with the routing results because it is not intended to just minimize distances. Geographically convenient shorter routes might actually be worse if such routes are associated with long waiting periods prior to harvest or an inappropriate delivery window for such a highly perishable crop. The VRPTW formulation thus focuses on realistic delivery timetables, considering perishability and suitability of drivers. The ranked pull workflow also offers an operating link between optimization and driver behavior: The mathematical solver makes the selection of an assignment, but the driver serves as the gate. Fig 6 illustrates how a single quotation ties these stages together end to end, populated with the real STCE point estimate and interval reported above and with the simulated settlement and routing figures reported in Table 2.




Fig. 6. Illustrative price-quotation and settlement interface, populated with the real model and simulation outputs reported above (UI mockup, not a live production screenshot).

The main contribution at the system level is thus in the conjunction of prediction and execution and not necessarily in the development of any specific algorithm. The forecasting layer generates a firm, bound price commitment; the escrow workflow takes a commitment from the forecasting layer, implements as an audited transaction state while the rule-based settlement engine adjusts the settlement without violating the firm commitment boundary, and VRPTW handles delivering the feasibility. SHAP acts as an interpretability layer for the price recommendations given, exposing the reasons behind them, while the rule-based settlement engine exposes the reasons behind every settlement adjustment by construction. The crop advisory module expands the system upstream by providing counselling to an expected market situation to support next-season crop production decisions. These are all elements that form an unbroken line of communication from pre-harvest planning right through to post delivery settlement.

The experimental realization has to be carried out using a normal development workstation or laptop. We recommend the Intel Core i7 or AMD Ryzen 7 processor, at least 16 GB RAM, NVIDIA RTX Series GPUs with at least 6 GB of VRAM for deep-learning experiments and for running Windows 11 or Ubuntu Linux for the reproducible configuration. The architecture can be made up of a lightweight front end for the PWA which is written in JavaScript/React, and a backend that is written in Python. The computational pipeline may utilize Python, PyTorch, scikit-learn, SHAP and Google OR-Tools. The final experiment in this paper is run on Python 3.10 with PyTorch 2.0.1, scikit-learn 1.2.2, SHAP 0.41.0 and Google OR-Tools 9.6, on an Intel Core i7 workstation with 16 GB RAM and an NVIDIA RTX-series GPU with 6 GB VRAM, running under Ubuntu Linux. The STCE forecaster was trained for 100 epochs using the Adam optimizer with an initial learning rate of 0.001, a batch size of 32, the mean-squared-error (MSE) loss function and a fixed random seed of 42. The chief source of market data is the website of the Government of India which contains information relating to the mandi-prices and is linked to AGMARKNET.

## V. Conclusion

AgriMart AI serves as a comprehensive, unified decision-support platform across the agricultural supply chain, integrating price forecasting, contractual pre-commitment, automated settlement, logistics routing, quality verification, and farmer advisories into a cohesive workflow. The foundational strength of the system lies in its capacity to transform standard price predictions into transparent, interpretable price bands that stakeholders can review and approve prior to shipment dispatch . The change is then implemented in the subsequent rule-based settlement mechanism, which lets the final market and transit conditions affect the final transaction and keeps the initial commitment boundary unchanged. Moreover, Perishability-aware VRPTW extends logistics to economic issues, and SHAP allows for greater transparency towards end users. The system is designed to work in a practical implementation using limited connectivity without any real financial infrastructure, but with simulated escrow. The manual quality verification, simulated data from the logistics are still key constraints. Some of the future work includes adding real payment gateways, adding automated computer-vision based quality grading of samples, adding more fleet data, adding federated learning across regions of the country and adding learning based routing. Bigger multi-market experiments and potential field tests would also be useful to validate economics benefits and user

acceptance. The framework, as a whole, supplies a secure base for tying predictive farming intelligence to secure execution of transactions and supply chain management for managing perishable goods.

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